

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

IManiyam Anushka(192525411)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Maniyam Anushka

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – CO2 SCENARIO-BASED (ANSWER KEY)

Course Name: Computer Networks Course Code: CSA07 06

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Question 1 – Smart Campus Subnetting (192.168.100.0/24)

Step-by-step working:

- /25 block size = $2^{(32-25)} = 128 \rightarrow$ range .0 to .127
- /26 block size = $2^{(32-26)} = 64 \rightarrow$ next range .128 to .191
- /27 block size = $2^{(32-27)} = 32 \rightarrow$ next range .192 to .223
- Remaining space in the /24 block = .224 to .255 (32 addresses unused)

ANSWER – School of Computing (/25)

Network Address	192.168.100.0
Broadcast Address	192.168.100.127
Host Range	192.168.100.1 – 192.168.100.126
Usable Hosts	$2^7 - 2 = 126$

ANSWER – School of Electronics (/26)

Network Address	192.168.100.128
Broadcast Address	192.168.100.191

Host Range	192.168.100.129 – 192.168.100.190
Usable Hosts	$2^6 - 2 = 62$

ANSWER — School of Mechanical Engineering (/27)

Network Address	192.168.100.192
Broadcast Address	192.168.100.223
Host Range	192.168.100.193 – 192.168.100.222
Usable Hosts	$2^5 - 2 = 30$
Remaining Unused Range	192.168.100.224 – 192.168.100.255

Question 2 — Software Company Subnetting (10.10.0.0/16)

Step-by-step working:

- /24 block size = 256 → range 10.10.0.0 to 10.10.0.255
- /25 block size = 128 → next range 10.10.1.0 to 10.10.1.127
- /26 block size = 64 → next range 10.10.1.128 to 10.10.1.191
- /27 block size = 32 → next range 10.10.1.192 to 10.10.1.223
- Remaining = 10.10.1.224 to 10.10.255.255 (rest of the /16 block)

ANSWER — Software Development (/24)

Network Address	10.10.0.0
Broadcast Address	10.10.0.255

Host Range	10.10.0.1 – 10.10.0.254
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ANSWER — Quality Assurance (/25)

Network Address	10.10.10
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Broadcast Address	10.10.127
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Host Range	10.10.11 – 10.10.126
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Usable Hosts	$2^7 - 2 = 126$
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ANSWER — Human Resources (/26)

Network Address	10.10.128
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Broadcast Address	10.10.191
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Host Range	10.10.129 – 10.10.190
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Usable Hosts	$2^6 - 2 = 62$
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ANSWER — Executive Management (/27)

Network Address	10.10.192
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Broadcast Address	10.10.223
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Host Range	10.10.193 – 10.10.222
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Usable Hosts	$2^5 - 2 = 30$
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Remaining Unused Range	10.10.1224 – 10.10.255.255
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Question 3 – Smart Hospital IPv6 Migration

Given: 2001:0db8:abcd:0000:0000:0000:0000 /64

Step-by-step working:

- Compression rule: remove leading zeros in each group, then replace the longest run of all-zero groups with ::
- Drop leading zeros: 2001:db8:abcd:0:0:0:0:0
- Replace consecutive zero groups → 2001:db8:abcd::
- Expansion reverses this: restore :: to full zero groups and pad each group to 4 hex digits
- /64 means the first 64 bits (4 groups) form the network prefix; remaining 64 bits are the host ID
- Total host addresses = 2^{64} (all 64 host bits available)

ANSWER – IPv6 Address Details

Compressed Notation	2001:db8:abcd::
Expanded Notation	2001:0db8:abcd:0000:0000:0000:0000:0000
Network Prefix (/64)	2001:0db8:abcd:0000
Host Portion	0000:0000:0000:0000 (last 64 bits)
Total Host Addresses	$2^{64} = 18,446,744,073,709,551,616$

Question 4 – ISP Routing Table Matching

Given: Routing entries 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24 | Packet destination:

192.168.2.45

Step-by-step working:

- 192.168.2.0/24 covers the range 192.168.2.0 – 192.168.2.255
- 192.168.2.45 falls inside this range → match found
- Router forwards the packet out of the interface connected to 192.168.2.0/24
- If no entry matched, the router would use the default route (0.0.0.0/0), typically pointing to the upstream ISP gateway

ANSWER — Routing Decision

Destination Subnet	192.168.2.0/24
Matching Routing Entry	192.168.2.0/24
Next Hop / Interface	Interface directly connected to 192.168.2.0/24
Default Route Used	0.0.0.0/0 (only if no entry matches)

Question 5 — Two Office LANs via Layer-3 Router

Given: Administration LAN 192.168.10.0/26 (Gateway .1) | Finance LAN 192.168.10.64/26 (Gateway .65)

Step-by-step working:

- /26 block size = 64 addresses
- Administration block: 192.168.10.0 – 192.168.10.63
- Finance block: 192.168.10.64 – 192.168.10.127
- The first usable IP in each LAN is reserved for the gateway, so suggested host IPs start right after it

ANSWER — Administration LAN (192.168.10.0/26)

Network Address	192.168.10.0
Broadcast Address	192.168.10.63
Host Range	192.168.10.1 – 192.168.10.62
Suggested Host IPs	192.168.10.2, 192.168.10.3, 192.168.10.4
Default Gateway	192.168.10.1

ANSWER — Finance LAN (192.168.10.64/26)

Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Host Range	192.168.10.65 – 192.168.10.126
Suggested Host IPs	192.168.10.66, 192.168.10.67, 192.168.10.68
Default Gateway	192.168.10.65